

A Survey of Quantum Machine Learning: Foundations, Algorithms, Frameworks, Data and Applications

FRANCESC RODRÍGUEZ-DÍAZ, Data Science and Big Data Lab, Universidad Pablo de Olavide, Seville, Spain

DAVID GUTIÉRREZ-AVILÉS, Department of Computer Science, University of Seville, Seville, Spain

ALICIA TRONCOSO, Data Science and Big Data Lab, Universidad Pablo de Olavide, Sevilla, Spain

FRANCISCO MARTÍNEZ-ÁLVAREZ, Data Science and Big Data Lab, Universidad Pablo de Olavide, Sevilla, Spain

Quantum machine learning combines quantum computing with machine learning to solve complex computational problems more efficiently than classical approaches. This survey provides an introduction to the foundations, algorithms, frameworks, data and applications of quantum machine learning, serving as a resource for researchers and practitioners. We begin by reviewing existing surveys to identify gaps that this work addresses, followed by a detailed discussion of the foundational principles of quantum mechanics and machine learning essential for quantum machine learning. Key algorithms are examined, highlighting their mechanisms, advantages, and applications across various domains. Current frameworks and platforms for implementing quantum machine learning algorithms are explored, emphasizing their unique features and suitability for different contexts. Existing quantum datasets for practical usage are also reported and commented on. This survey also reviews over 135 articles, categorized into theoretical and practical contributions, to identify key advances, limitations, and application areas within quantum machine learning. Critical challenges such as hardware limitations, error rates, and scalability are analyzed to detect the obstacles that must be addressed for practical deployment. By synthesizing these elements into a structured overview, this survey aims at serving as both an introduction and a guide for advancing research and development in this disruptive field.

CCS Concepts: • **Computing methodologies** → **Machine learning**; **Artificial intelligence**; • **Mathematics of computing** → **Information theory**; • **Applied computing**; • **Hardware** → **Emerging technologies**; • **General and reference** → **Surveys and overviews**;

Additional Key Words and Phrases: Quantum machine learning, quantum foundations, quantum algorithms, quantum computing frameworks, quantum datasets, quantum applications

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Authors' Contact Information: Francesc Rodríguez-Díaz, Data Science and Big Data Lab, Universidad Pablo de Olavide, Seville, Spain; e-mail: froddia@upo.es; David Gutiérrez-Avilés, Department of Computer Science, University of Seville, Seville, Spain; e-mail: dgutierrez3@us.es; Alicia Troncoso, Data Science and Big Data Lab, Universidad Pablo de Olavide, Sevilla, Spain; e-mail: atrolor@upo.es; Francisco Martínez-álvarez, Data Science and Big Data Lab, Universidad Pablo de Olavide, Sevilla, Spain; e-mail: fmaralv@upo.es.



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1 Introduction

The intersection of quantum computing and machine learning has led to an emerging field known as **Quantum Machine Learning (QML)**, promising to fundamentally transform how we approach complex information processing and decision-making issues [10, 42]. QML combines the foundational principles of quantum mechanics with advanced machine learning techniques, making use of the properties of quantum systems, such as superposition, entanglement and quantum parallelism, to achieve computational capabilities beyond classical limitations [107].

As we navigate the era of quantum computing, the integration of quantum mechanics with machine learning has become a pivotal research area, attracting significant attention from both academia and industry. This interdisciplinary domain can potentially revolutionize various sectors, including cryptography, optimization, artificial intelligence, and data science, by offering novel solutions to problems that were once considered intractable [18].

Specifically, this review analyzes the current state of QML, highlighting recent advances, practical challenges, and emerging application areas. QML explores computational advantages in solving complex problems that are often intractable for classical systems. Despite its potential, the existing literature remains fragmented, since many surveys focus narrowly on either theoretical principles or specific algorithmic implementations. To address this, we present a structured overview that unifies the theoretical foundations of QML, algorithmic developments, hybrid architectures, implementation frameworks, available datasets, and real-world use cases. Hence, this work aims at serving as both an accessible introduction and a complete reference for researchers and practitioners.

We also examine the technical limitations of deploying QML algorithms on current quantum hardware, including challenges related to qubit decoherence, quantum error correction, and scalability. Alongside these constraints, we emphasize the notable milestones already achieved in the field and identify promising future directions. Finally, this review provides a perspective on the opportunities and limitations of QML.

Given the above, our main contributions are as follows:

- (1) We provide an integration of quantum foundations and machine learning techniques. In particular, this survey thoroughly consolidates foundational quantum computing concepts, including quantum superposition, entanglement, coherence, parallelism, and variational quantum circuits, with advanced machine learning techniques.
- (2) A detailed examination and taxonomy of quantum algorithms in machine learning is provided, categorizing them into foundational quantum algorithms, QML algorithms, quantum deep learning, quantum reinforcement learning and quantum optimization algorithms. This analysis highlights their mechanisms, theoretical advantages and current limitations.
- (3) A systematic identification and categorization of the critical challenges faced in the practical implementation of QML are presented. Specifically, the survey addresses hardware constraints, algorithmic scalability issues, data encoding complexities, and interpretability of quantum models.
- (4) An overview of future research directions and emerging trends in QML is articulated. The survey discusses promising areas, including hybrid quantum-classical algorithms, advanced error mitigation techniques, novel quantum architectures, and potential breakthroughs required to transition QML from theoretical promise to mainstream technological integration.

The rest of the article is structured as follows: Section 2 discusses related work, providing an overview of other existing surveys on QML, categorized into general overviews, application-specific surveys, and algorithm-focused surveys. Section 3 outlines the foundations of quantum computing, introducing essential principles and key quantum algorithms. Section 4 explores QML and foundational quantum algorithms, outlining prominent approaches. Section 5 introduces hybrid architectures, highlighting their current potential. In Section 6, we present an overview of frameworks, cloud integration and existing quantum datasets and repositories. Section 7 provides an analysis of current research in QML, categorizing proposals into theoretical and practical approaches, literature reviews, and direct applications. It highlights advancements in algorithm design, implementation challenges, and the potential of QML across diverse sectors. Section 8 explores the main challenges and future directions in QML, including hardware constraints, algorithmic complexity, data encoding issues, and ethical concerns, and discusses potential solutions such as hybrid approaches and noise-resilient models. Section 9 presents the conclusions, highlighting QML's progress, potential, and the need for continued interdisciplinary efforts to overcome current challenges and enable practical applications. Supplementary material can be found in the appendices, with Appendix A containing tables, Appendix B presenting figures, and Appendix C providing the pseudocode for the algorithms discussed.

2 Related Works: Other Existing Surveys

Over the years, several surveys have been conducted to explore the advancements, applications, challenges, and future directions of QML. This section discusses previous surveys on this topic, highlighting their contributions, strengths, and limitations. Thus, Section 2.1 presents general overviews of QML, Section 2.2 reviews application-specific surveys, Section 2.3 focuses on algorithm-centered studies, and Section 2.4 examines surveys addressing challenges and future perspectives in the field.

2.1 General Overviews of QML

One of the foundational surveys in this area can be found in [53], which provides a comparative analysis of quantum and classical machine learning approaches. This survey delves into fundamental quantum algorithms and examines scenarios where quantum methods may offer computational advantages over classical techniques. It effectively introduces readers to the potential speedups and efficiencies that quantum algorithms can bring to machine learning tasks. However, the survey primarily focuses on theoretical aspects and lacks detailed discussions on practical implementations and experimental validations.

Similarly, [119] offers an overview of various quantum algorithms utilized in machine learning, such as quantum support vector machines, quantum principal component analysis, and quantum neural networks. The survey discusses potential applications across multiple domains, providing a broad perspective on the possibilities within QML. Readers without a strong background in quantum computing might find some sections challenging due to the technical complexity involved.

The work in [33] bridges the gap between theoretical algorithms and practical applications by reviewing key QML algorithms and presenting case studies that demonstrate their real-world implementations.

In [15], the authors highlight emerging trends and recent developments in QML, including novel algorithms, improvements in quantum hardware, and the integration of quantum computing with existing machine learning frameworks. This survey provides up-to-date information and identifies potential future research directions.

Lastly, [93] conducts a systematic and scientometric analysis of quantum-inspired metaheuristics, which take advantage of quantum principles for optimization problems. Although it focuses primarily on optimization, the results provide insight into the broader implications of quantum-inspired methods within machine learning.

2.2 Application-specific Surveys

In the healthcare domain, [73] examines how QML can address complex biomedical challenges, such as genomic analysis, medical imaging, and drug discovery. The survey analyzes the potential of QML to improve diagnostic accuracy and facilitate personalized medicine. However, it may not extensively cover the technical implementation details of quantum algorithms. García and Ullah [117] explore the transformative potential of QML in healthcare systems. The survey focuses more on high-level concepts and emerging perspectives than on providing detailed technical analyses, which may limit its utility to those seeking a deep understanding of specific algorithms or implementation challenges. The review by [129] provides an overview of the intersection between quantum-enabled algorithms and medical image processing, highlighting applications in disease diagnosis, segmentation, and security. It identifies key challenges, such as the immaturity of quantum hardware and the need for more advanced quantum algorithms.

In the telecommunications field, [114] investigates the integration of machine learning and quantum computing within the context of next-generation communication networks. The survey addresses the role of QML in the advancement of communication technologies, discussing practical challenges such as network optimization, resource allocation, and security considerations. The review in [34] focuses on quantum-inspired machine learning techniques specifically adapted for 6G networks. The survey emphasizes foundational concepts, security aspects, and resource management challenges. Explores how quantum principles can enhance classical machine learning methods to meet the demands of future communication systems. The survey primarily offers theoretical insights rather than guidance on practical implementation.

In the context of metaverse investments, [61] introduces a novel decision-making framework to improve the performance of metaverse projects. This study highlights key performance indicators such as organizational effectiveness, technological advancements, and regulatory frameworks, highlighting their critical role in investment success. The application focus underscores the practical use of quantum-inspired fuzzy logic in prioritizing criteria and ranking alternatives. Kou and Lu [62] present a review of emerging financial technologies, focusing on the rapid evolution and application of FinTech in financial services. It identifies computational efficiency and security as key drivers for future research, suggesting potential intersections with quantum-enhanced methods.

In the optimization domain, [36] provides a detailed review of quantum metaheuristic algorithms, exploring their integration of quantum computing principles to enhance global optimization capabilities. It highlights the benefits of quantum computing in improving convergence rates and balancing exploration and exploitation in search spaces.

2.3 Algorithm-focused Surveys

The authors in [32] provide an in-depth examination of the Harrow-Hassidim-Lloyd (HHL) algorithm for solving linear systems of equations. The survey discusses the theoretical foundations of the HHL algorithm and its applications in QML, such as regression analysis and principal component analysis. It offers valuable insights for readers with a strong background in quantum computing but with high technical complexity.

Zhang et al. [116] explore the development of quantum neural networks, specifically, architectures, training methods, and their potential advantages in modeling complex data distributions. However, it may not cover other applications of quantum neural networks beyond generative models.

Schuld and Pretuccione [107] review quantum algorithms specifically designed for supervised learning tasks, such as classification and regression. The survey focuses on algorithms like quantum support vector machines and quantum K-nearest neighbors.

The impact of quantum noise on machine learning algorithms and exploring methods to mitigate these effects is addressed in [71]. The survey focuses on the practical challenges of implementing

QML on **noisy intermediate-scale quantum (NISQ)** hardware, discussing error mitigation techniques and the design of noise-resilient algorithms. Its high technical complexity and specialized focus may make it less accessible to readers without experience in quantum error correction and hardware-specific issues.

2.4 Challenges and Future Perspective Surveys

Biamonte et al. [10] identify the current obstacles facing QML, including hardware limitations, scalability issues, and the need for efficient algorithmic designs. The survey discusses opportunities for future research and provides a balanced view of hurdles and potential advances in the field. However, it may lack detailed solutions or concrete strategies to overcome these challenges, and some discussions may be speculative rather than offering actionable guidance. However, it is one of the most prominent surveys in this field and should be read by any novel researcher.

The work in [55] offers an extensive review of various QML algorithms and quantum deep learning techniques. Theoretical developments and practical challenges are addressed, discussing algorithmic efficiency, hardware considerations, and potential applications across different domains. Although the survey is comprehensive, the breadth of topics covered might be overwhelming to some readers, and a prior understanding of quantum computing principles may be necessary to fully appreciate the content.

Finally, the survey [131] discusses important topics in quantum computing and communications that are relevant to the development of QML. Provides context on how QML fits within the broader landscape of quantum technologies, addressing foundational concepts, infrastructure challenges, and potential impacts on various industries. Although offering valuable background information, the survey is not focused exclusively on machine learning algorithms, which may limit its usefulness for readers seeking detailed QML-specific insights.

Existing surveys on QML help clarify the field's progress by addressing algorithms, applications, challenges, and future directions. General surveys offer accessible overviews for non-expert audiences, while specialized studies focus on specific domains or techniques. Continued consolidation and periodic literature updates are necessary to maintain the relevance and usefulness of QML resources for researchers and practitioners. Table 1 (see Appendix A) summarizes all the surveys reviewed, including application area, highlights, and limitations.

3 Foundations of Quantum Computing

This section provides the essential foundations of quantum computing relevant to QML. Section 3.1 introduces core concepts such as qubits, quantum gates, and fundamental quantum operations. Section 3.2 extends this by describing quantum circuits and variational quantum circuits, which integrate classical and quantum resources for optimization and learning tasks. Section 3.3 discusses key phenomena including quantum coherence and parallelism, while Section 3.4 addresses quantum noise and error mitigation strategies critical for reliable quantum computation.

3.1 Basic Quantum Information Units and Operations

A qubit, or quantum bit, is the fundamental unit of quantum information in quantum computing. Unlike classical bits, which can only be in a state of either 0 or 1, qubits utilize quantum mechanics to exist in a superposition of both states simultaneously. This unique property allows quantum computers to perform multiple calculations at once, potentially solving certain problems much more efficiently than classical computers. The mathematical representation of a qubit is shown in Equation (1):

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad (1)$$

where α and β are complex probability amplitudes. Additionally, α and β must satisfy the constraint given in Equation (2):

$$|\alpha|^2 + |\beta|^2 = 1. \quad (2)$$

This equation indicates that the total of the squared probabilities must sum up to one.

The quantum states of a qubit system can be represented in the Hilbert space, a specialized vector space endowed with an inner product, represented by $\langle\phi|\psi\rangle$. This concept is essential for studying vector spaces in quantum mechanics, offering a mathematical structure to describe the properties and dynamics of quantum systems. This space can be illustrated using the Bloch sphere, as shown in Figure 1 (see Appendix B). The Bloch sphere is a visual tool that allows for the representation of all possible pure states of a qubit on the surface of a unit sphere. However, not all quantum states can be represented in the Bloch sphere, particularly when dealing with systems comprising more than one qubit or with mixed states. Mixed states, for instance, necessitate a formalism that goes beyond this simplified geometric representation.

On the Bloch sphere:

- The Z-axis defines the basis states, placing the state $|0\rangle$ at the north pole and the state $|1\rangle$ at the south pole.
- The X and Y axes depict superposition's of these basis states with varying relative phases, enabling the visualization of all pure states of a qubit as points on the sphere's surface.

Each pure state of a qubit can be described by a linear combination of the basis states $|0\rangle$ and $|1\rangle$, and is representable as a single point on the Bloch sphere. This representation is beneficial for understanding essential concepts in quantum information theory and quantum computing, such as superposition and quantum operations. Mathematically, a pure state $|\psi\rangle$ is expressed as Equation (1). The density matrix for a pure state is defined as $\rho = |\psi\rangle\langle\psi|$. For example, if $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, the density matrix is

$$\rho = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}. \quad (3)$$

The main properties of the density matrix are

- (1) Hermiticity: $\rho = \rho^\dagger$, which means the density matrix is equal to its conjugate transpose. This ensures that the expected values of observable are real.
- (2) Traceability: $\text{Tr}(\rho) = 1$, which guarantees that the total probability is 1.
- (3) Positivity: ρ is a positive semi definite operator, which means that for any state vector $|\phi\rangle$, $\langle\phi|\rho|\phi\rangle \geq 0$. This ensures that probabilities are non negative.
- (4) Eigenvalues: The eigenvalues of ρ are in the range $[0, 1]$ and their sum is equal to 1, reflecting the normalization of probabilities.

For the above pure state, the trace of the density matrix, which is the sum of its diagonal elements, is

$$\text{Tr}(\rho) = \frac{1}{2}(1 + 1) = 1. \quad (4)$$

The trace of the density matrix must always be equal to 1, ensuring that the total probability of the quantum system is 1.

The main properties of the trace are

- (1) Linearity: The trace is a linear operation. For any two matrices A and B and any complex number α ,

$$\begin{aligned} \text{Tr}(A + B) &= \text{Tr}(A) + \text{Tr}(B) \\ \text{and } \text{Tr}(\alpha A) &= \alpha \text{Tr}(A) \end{aligned} \quad (5)$$

- (2) Cyclicity: The trace is invariant under the cyclic permutation of the product of matrices. For any two matrices A and B ,

$$\text{Tr}(AB) = \text{Tr}(BA). \quad (6)$$

- (3) Invariance: The trace of a matrix is invariant under unitary transformations. If U is a unitary matrix ($U^\dagger U = I$), then for any matrix A ,

$$\text{Tr}(U A U^\dagger) = \text{Tr}(A). \quad (7)$$

- (4) Invariance under change of basis: The trace of a matrix is independent of the basis in which it is expressed.

However, a mixed state is a statistical mixture of several pure states. Instead of being described by a single state vector, a mixed state is represented by a density matrix that combines various possible configurations of the system, each with a certain probability. Mixed states allow us to represent quantum systems that are not in a pure state due to uncertainty or incomplete information about the system. Mathematically, a mixed state is represented by a density matrix ρ , which is a convex combination of pure state projectors:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \quad (8)$$

where p_i are the probabilities (with $\sum_i p_i = 1$). For example, consider a qubit that is in a mixed state with a probability p of being in the state $|0\rangle$ and a probability $1 - p$ of being in the state $|1\rangle$. The density matrix for this mixed state can be written as follows:

$$\rho = p|0\rangle\langle 0| + (1 - p)|1\rangle\langle 1|. \quad (9)$$

Then, the density matrix ρ can be written as

$$\begin{aligned} \rho &= p \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + (1 - p) \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} p & 0 \\ 0 & 1 - p \end{pmatrix}. \end{aligned} \quad (10)$$

For this mixed state, the trace of the density matrix is as follows:

$$\text{Tr}(\rho) = p + (1 - p) = 1. \quad (11)$$

3.2 Quantum Circuits

To exploit the potential of quantum computing, classical data, typically represented as binary bits, must first be encoded into quantum states, known as qubits. This process, referred to as data encoding, embedding, or feature mapping, is a fundamental step in developing quantum and hybrid quantum-classical models. While feature mapping also exists in classical machine learning, encoding data into quantum states is specific to quantum computing and enables the use of quantum phenomena such as superposition and entanglement.

A quantum circuit visually represents a sequence of quantum operations performed on qubits. These operations range from simple actions, like applying gates to individual qubits, to more complex procedures involving multiple qubits. Quantum circuits are similar to their classical counterparts in digital electronics but function based on the principles of quantum mechanics. In Figure 2 (see Appendix B), we can observe an example of a quantum circuit. The circuit is composed of a series of channels, with each channel representing a distinct qubit, and a series of gates, specifically Hadamard, CNOT, Z and measure gates, the latter represented by a counter. Table 2 (see Appendix A) presents the most common gates along with a brief description of each.

A particularly important class of quantum circuits in the context of quantum machine learning is the **Variational Quantum Circuit (VQC)**. The key idea behind VQCs is to construct a parameterized quantum circuit with fixed architecture but adjustable parameters (e.g., rotation angles), which are optimized to minimize a classical cost function.

A VQC typically consists of three main stages. First, classical input data are encoded into quantum states using an embedding circuit or feature map. Second, a sequence of parameterized quantum gates is applied to the qubits, forming the trainable part of the circuit. These gates depend on real-world parameters that are tuned during training. Finally, quantum measurements are performed, and the resulting expectation values are used to evaluate a classical cost function, which is subsequently minimized using classical optimization algorithms.

This hybrid process exploits quantum effects such as superposition and entanglement, while outsourcing parameter optimization to classical processors. Therefore, VQC is particularly well-suited to NISQ devices, which have limited coherence time and a limited number of qubits. VQCs serve as a fundamental component in several algorithms presented in Section 4.

3.3 Quantum Coherence and Parallelism

The power of quantum computing comes from uniquely non-classical phenomena, especially quantum coherence and quantum parallelism. These two features are not only foundational to quantum mechanics but also essential to the operation and advantage of quantum algorithms over their classical counterparts.

On the one hand, quantum coherence is the ability of a quantum system to maintain phase relationships between different quantum states in superposition. This property allows quantum computers to manipulate and interfere with these states in a controlled manner. Coherence is a necessary condition for any quantum computation to produce meaningful results. Once coherence is lost, the system behaves classically. Mathematically, quantum coherence is encoded in the off-diagonal elements of the density matrix ρ of a quantum system. Given a quantum state expressed in a particular orthonormal basis $|i\rangle$, the density matrix takes the form of Equation (12).

$$\rho = \sum_{i,j} \rho_{i,j} |i\rangle\langle j|. \quad (12)$$

The diagonal elements ρ_{ii} represent the populations (i.e., probabilities) of finding the system in the basis state $|i\rangle$, while the off-diagonal elements ρ_{ij} for $i \neq j$ represent coherences, i.e., the quantum correlations or phase relationships between the states $|i\rangle$ and $|j\rangle$. These terms are essential for the manifestation of interference effects in quantum algorithms. Preserving coherence is crucial to enable phenomena such as entanglement and interference, fundamental resources in quantum information processing.

In quantum algorithms, coherence enables interference patterns that amplify correct solutions while suppressing incorrect ones, a principle exploited in algorithms such as Grover's search and quantum phase estimation [135]. Without coherence, the constructive and destructive interference necessary for quantum speedup cannot occur. Moreover, in hybrid quantum-classical models like VQC, the ability to maintain coherence throughout the circuit execution directly impacts the fidelity and optimization quality of the learning process [95].

Coherence is also essential beyond algorithms, especially in quantum communication and sensing technologies [92]. In quantum key distribution, for instance, preserving coherence ensures secure state transmission and enables detection of eavesdropping [81].

On the other hand, quantum parallelism refers to the capability of a quantum system to evaluate a function on multiple inputs simultaneously by exploiting the principle of superposition. Formally, given a function $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$ and its associated unitary transformation U_f , defined by

$U_f|x\rangle|0\rangle = |x\rangle|f(x)\rangle$, a quantum computer can operate on a superposition of inputs $\sum_x \alpha_x|x\rangle$, producing the state $\sum_x \alpha_x|x\rangle|f(x)\rangle$. This means that all evaluations of $f(x)$ are encoded within the resulting quantum state after a single application of U_f . Although individual function values cannot be directly extracted from this superposition, quantum algorithms exploit interference and measurement to extract global properties of f , providing computational advantages over classical approaches.

The importance of quantum parallelism in practical QML is illustrated in [124], in which the authors investigate its role in enabling data parallelism in QNN. While classical neural networks rely heavily on data parallelism to improve training efficiency, their quantum analog has remained underexplored. This study demonstrates that quantum parallelism, achieved through quantum superposition and quantum entanglement, can be effectively applied to process multiple data points simultaneously.

Further details on the nature of quantum parallelism can be found in [75], which provides a conceptual and analytical comparison between quantum parallel and classical computing.

3.4 Quantum Noise and Error Mitigation

As quantum computers evolve from theoretical constructs to physical implementations, they face a fundamental challenge known as quantum noise. Quantum noise is a type of error that can occur in various ways during the execution of a quantum system. There are several types of noise, but due to the nature of the survey, we will only mention the following three: Coherent noise [17], projection noise [49], and incoherent noise [12].

Coherent noise is a type of noise that occurs during the execution of quantum gates on a qubit. This type of noise is predictable and can be mitigated through various techniques. For example, let us consider an X gate, which performs a flip on the qubit state such that if the state is $|0\rangle$, it changes to $|1\rangle$, and vice versa. If we were to execute this operation on a simulator and represent this behavior on the Bloch sphere, we would see what is illustrated in Figure 3(a) (see Appendix B).

However, if we were to execute the same operation in a real environment and represent the results in the Bloch sphere, we would observe that the result does not exactly yield the state $|1\rangle$, but rather has shifted slightly more than expected (Figure 3(b), see Appendix B). One way to mitigate this type of noise is by incorporating noise gates into the circuit, which essentially act as a correction factor for the error. Figure 4 (see Appendix B) is a quantum circuit that includes such a gate for the mitigation of this type of error. In this case, the error gate shown depends on a parameter ϵ , which indicates the deviation of the quantum gate operation from its ideal behavior.

The next type of error is projection noise. This error refers to the inaccuracies that occur during the measurement process of quantum states. When we measure a qubit, it provides a probability that the qubit is either one or zero. This probability is defined as follows:

$$X = 0 : \quad p(X = 0) = |\langle 0|\psi\rangle|^2, \quad (13)$$

$$X = 1 : \quad p(X = 1) = |\langle 1|\psi\rangle|^2. \quad (14)$$

The sum of these probabilities must always equal 1, such that:

$$p(X = 0) + p(X = 1) = 1. \quad (15)$$

Noise can affect these probabilities such that their sum does not equal exactly 1, and it can even increase their variance and standard deviation.

A possible solution to this error is to perform M shots of the circuit, where the M outputs follow an **independent and identically distributed (IID)** distribution. The term “independent” refers

to each outcome that does not influence any other outcome. The result of one measurement has no impact on the results of other measurements. This implies that the noise or error that affects one shot is not correlated with the noise or error affecting another shot. The term “identically distributed” means that each outcome follows the same probability distribution. This ensures that the statistical properties (such as mean, variance, etc.) of the outcomes are consistent across all measurements. Thus, we would have:

$$X_1, X_2, \dots, X_M \sim \text{Pr}[X = x]. \quad (16)$$

In addition, note that

$$X_1, X_2, \dots, X_M \in \{0, 1\}. \quad (17)$$

Finally, the probability S that we obtain after the M shots is computed as follows:

$$S = \frac{1}{M} \sum_{m=1}^M X_m. \quad (18)$$

The final type of noise we will examine is incoherent noise, which encompasses random and unpredictable errors, such as decoherence and dissipation. Decoherence occurs when a quantum system interacts with its environment, causing the loss of quantum coherence and reducing the system’s ability to maintain superposition states. Dissipation, on the other hand, refers to the process by which the quantum system loses its quantum properties altogether, transitioning to a more classical behavior.

To model and address incoherent noise, several approaches are commonly used. X bit-flip noise [104] represents errors in which a qubit’s state is randomly flipped between $|0\rangle$ and $|1\rangle$, simulating physical disturbances in the system. Phase noise [72] captures errors that affect the relative phase between the quantum states of a qubit, which can distort interference patterns crucial to quantum operators. Depolarizing noise [120] is a general model in which a qubit’s state is replaced by a completely mixed state with a certain probability, effectively erasing the information in the qubit.

For a better understanding of all these types of noise and errors, refer to [20], where the authors present all types of errors and noise that can occur and the methods to address them.

4 Quantum Algorithms

This section provides an overview of key algorithmic categories in quantum computing, including foundational methods (Section 4.1), learning algorithms (Section 4.2), deep learning techniques (Section 4.3), reinforcement learning (Section 4.4), and optimization approaches (Section 4.5).

4.1 Foundational Quantum Algorithms

Quantum algorithms exploit the principles of quantum mechanics to solve problems more efficiently than classical algorithms. They exploit phenomena such as superposition and entanglement to achieve significant computational speedups. This subsection explores some of the most prominent fundamental quantum algorithms.

4.1.1 Grover’s Algorithm. This is one of the most fundamental algorithms in quantum computing. This algorithm fully exploits the concept of superposition to perform a search over a set of N unstructured data elements. Any search problem can be expressed as a function $f(x)$, where x is an element of the set to be searched. If the element x is a solution, then $f(x) = 1$; otherwise, if x is not a solution, $f(x) = 0$. Classically, in the worst-case scenario, we would have to evaluate $f(x)$ a total of $N - 1$ times, where $N = 2^n$. This approach involves testing all possibilities. In contrast, Grover’s algorithm solves this problem with quadratic speedup, requiring only $\sqrt{N}$ evaluations, compared to N .

To achieve this speedup, Grover's algorithm defines a matrix called the oracle, which must act on all states $|x\rangle$. The matrix is defined as follows:

$$U_f|x\rangle = (-1)^{f(x)}|x\rangle. \quad (19)$$

The function indicates that if x is not a solution, it will have no effect on the state. However, if x is a solution, we have the following:

$$U_f|x\rangle = -|x\rangle. \quad (20)$$

With this, the matrix corresponds to performing a reflection in its amplitude on the marked state within the set of N elements. Essentially, the function highlights one state over the others by modifying the amplitude of that state. As a result, the amplitude of the remaining states decreases, and upon measurement, the algorithm will return the correct element with a probability close to 100%.

One of the key applications of Grover's algorithm is in cryptanalysis. It can be used to attack symmetric cryptographic primitives like block ciphers and hash functions. For instance, Grover's algorithm can find the secret key of a block cipher in $\theta(\sqrt{N})$ time, where N is the size of the key space. This means that Grover's algorithm can effectively halve the security level of symmetric cryptographic algorithms, posing significant implications for cybersecurity.

Another significant application of Grover's algorithm is in database search. In scenarios where data is stored in an unstructured manner and needs to be searched for a specific entry, Grover's algorithm provides an efficient way to perform such searches. This can be crucial for large-scale data analysis, where classical search methods would be too slow to be practical. Grover's algorithm's ability to search through an unsorted database or list of items in $\theta(\sqrt{N})$ time makes it an invaluable tool for improving the efficiency of data retrieval processes.

Finally, its pseudocode can be found in Algorithm 1 (see Appendix C).

4.1.2 Bernstein-Vazirani Algorithm. This is a fundamental algorithm in quantum computing that addresses a specific problem known as the Bernstein-Vazirani problem. This problem is a variant of the hidden function problem. The task is as follows: we are given a function f defined as $f(x) = s \cdot x \bmod 2$, where s is a hidden n -bit binary string and x is an n -bit binary string. Here, $\bmod 2$ (modulo 2) means that the result of the dot product $s \cdot x$ is taken modulo 2, which ensures the result is either 0 or 1. The goal is to determine the string s using the fewest possible evaluations of the function f .

Mathematically, the problem is defined as follows: The function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ is given by $f(x) = s \cdot x \bmod 2$. Here, $s = (s_1, s_2, \dots, s_n)$ is the hidden binary string we want to discover. The dot product $s \cdot x$ is the sum $s_1x_1 + s_2x_2 + \dots + s_nx_n$ calculated in base on modulo 2. In a classical algorithm, solving this problem would require n evaluations of the function f in the worst case, as each evaluation provides information about only one bit of s . The Bernstein-Vazirani algorithm efficiently solves this problem using quantum computing.

The Bernstein-Vazirani algorithm solves the problem with a single evaluation of the function f using the quantum oracle, representing an exponential improvement compared to the n steps required in a classical algorithm. The complexity of the quantum algorithm is $\theta(1)$ in terms of evaluations of f .

This algorithm can be used to diagnose errors in quantum circuits, as its straightforward implementation and ability to detect specific patterns in Boolean functions make it a useful tool for validating the correct operation of quantum gates. Additionally, this algorithm has applications in quantum cryptography, particularly in constructing protocols that require the detection of specific patterns or the verification of secret identities efficiently. The ability to discover a hidden string in a single step can be utilized to strengthen certain quantum cryptographic schemes.

Finally, its pseudocode can be found in Algorithm 2 (see Appendix C).

4.1.3 Shor's Factoring Algorithm. This is a quantum algorithm developed in 1994 by Peter Shor for finding the prime factors of an integer. It is one of the few quantum algorithms with significant potential applications and strong evidence of a superpolynomial speedup compared to the best-known classical algorithms. However, factoring numbers of practical importance requires more qubits than will be available in the near future, and the correction of quantum errors adds further complexity.

Shor also proposed algorithms for solving the discrete logarithm problem and the period-finding problem. Generally, "Shor's algorithm" refers to the factoring algorithm, but it may also encompass any of these methods.

On a quantum computer, Shor's algorithm can factor a number N in polynomial time, specifically $\theta((\log N)^2(\log(\log N))(\log(\log(\log N))))$ with fast multiplication, or even $\theta((\log N)^2(\log(\log N)))$ using the fastest known multiplication algorithm. This is significantly faster than the general number field sieve, the most efficient classical algorithm.

Cryptographic security is one of the primary applications of Shor's algorithm. The algorithm can potentially break systems like RSA, which rely on the difficulty of factoring large numbers. This vulnerability has driven significant research and development in post-quantum cryptography, aiming at creating cryptographic systems that remain secure against quantum attacks.

Additionally, the principles behind Shor's algorithm can be applied to other optimization problems that benefit from the speedup provided by quantum computing. The ability to solve such problems more efficiently could have wide-ranging impacts across various fields that require complex computations and optimizations.

Finally, implementing Shor's algorithm on quantum computers serves as a crucial test-bed for refining error correction techniques and improving the stability of qubits. This research is essential for advancing quantum computing technology and realizing its full potential in solving practical and theoretical problems.

Finally, its pseudocode can be found in Algorithm 3 (see Appendix C).

4.1.4 Quantum Fourier Transform Algorithm. The **Quantum Fourier Transform (QFT)** is a quantum algorithm that performs the discrete Fourier transform on the amplitudes of quantum states. It is a key component in many quantum algorithms and serves as the cornerstone of quantum computing applications involving periodicity and interference. The QFT operates exponentially faster than its classical counterpart, achieving a complexity of $\theta(n^2)$ for n qubits compared to $\theta(n^{2n})$ in the classical case. This speedup is critical in algorithms like Shor's algorithm (described in the previous section).

Mathematically, the QFT maps a quantum state $|x\rangle$ to a superposition of states weighted by Fourier coefficients. For an n -qubit input, the transformation is given by Equation (21):

$$\text{QFT}_n |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} e^{2\pi i \frac{xy}{2^n}} |y\rangle, \quad (21)$$

where x and y are integers between 0 and $2^n - 1$, $|x\rangle$ and $|y\rangle$ represent computational basis states. The factor $\frac{1}{\sqrt{2^n}}$ ensures the transformation is unitary, maintaining the normalization of quantum states.

The algorithm encodes the frequency spectrum of the input amplitudes in the quantum state, enabling further operations to extract meaningful results.

The QFT has a wide range of applications in quantum computing. In Shor's algorithm, it is a fundamental component for determining the periodicity of functions, enabling integer factorization, and solving the discrete logarithm problem. In phase estimation, the QFT is used to

extract eigenvalues of unitary operators, which is a key technique in algorithms such as Grover's search and quantum simulations. In addition, it is applied in signal processing, facilitating efficient analysis of quantum states in the frequency domain, with promising implications for quantum cryptography and optimization problems.

Despite its theoretical efficiency, QFT faces several practical challenges. The major issue is its sensitivity to errors, as controlled phase-shift gates require high precision, making them susceptible to noise and decoherence in current quantum systems. Moreover, qubit connectivity can pose significant obstacles, particularly in quantum hardware with limited inter-qubit interactions, as the QFT demands entanglement between distant qubits. These limitations complicate its implementation on NISQ devices, driving research into approximate versions and error mitigation strategies to enhance its practicality.

Finally, its pseudocode can be found in Algorithm 4 (see Appendix C).

4.2 Quantum Machine Learning Algorithms

QML algorithms combine quantum computing principles with traditional machine learning techniques to enhance performance and efficiency. These algorithms hold the promise of solving complex problems in high-dimensional spaces more effectively than classical methods. The following are the algorithms that have been successfully implemented in their quantum versions and are among the most commonly used: **Variational Quantum Classification Algorithm (VQCA)**, **Quantum Support Vector Machines (QSVM)**, and **Quantum Neural Networks (QNN)**.

4.2.1 Variational Quantum Classification Algorithm. VQCA is a widespread algorithm in QML, enabling quantum systems to perform classification tasks. This algorithm combines quantum circuits with classical optimization techniques. It begins with a parameterized quantum circuit, where the parameters of the gates can be adjusted. This circuit encodes input data into quantum states, and the parameters are iteratively optimized to minimize the discrepancy between predicted and actual labels. Classical optimization methods, guided by the quantum gradients of a defined cost function, adjust these parameters to enhance the model's accuracy.

Variational algorithms consist of several elements that can be combined and optimized according to advancements in algorithms, software, and hardware. Among these elements are a cost function that represents a specific problem with certain parameters, an initial structure that defines the search space with these parameters, and an optimizer that iteratively explores this space. In each iteration, the optimizer calculates the current cost with the existing parameters and determines the values these parameters should have for the next iteration. This process is repeated until the optimizer converges on an optimal solution. The hybrid nature of these algorithms arises from the fact that the cost functions are evaluated using quantum resources, while the optimization is carried out using classical resources.

The first step of the algorithm is to initialize the quantum computer in a default state $|0\rangle$. Then, this state is transformed into a desired (non-parametrized) state $|\rho\rangle$, which is referred to as the reference state. This transformation is achieved by applying a reference operator U_R to the default state, such that $U_R|0\rangle = |\rho\rangle$.

The second step of the process involves preparing the ansatz, which is an essential initial configuration for variational optimization. An ansatz is a parameterized quantum circuit designed to explore a subset of quantum states that may approximate the solution. This ansatz is mathematically represented as $U_A(\theta)|0\rangle$. This step begins by defining a variational form $U_V(\theta)$, representing a collection of parameterized states that the variational algorithm will explore. The ansatz combines a reference state with this variational form to create a parameterized quantum circuit. Specifically, the ansatz transforms the default state $|0\rangle$ into the target state $|\psi(\theta)\rangle$ by applying a variational form

over a reference operator. The equation representing this process is as follows:

$$\begin{aligned}
 |0\rangle &\xrightarrow{U_R} U_R|0\rangle = |\rho\rangle \xrightarrow{U_V(\theta)} U_A(\theta)|0\rangle \\
 &= U_V(\theta)U_R|0\rangle \\
 &= U_V(\theta)|\rho\rangle \\
 &= |\psi(\theta)\rangle.
 \end{aligned} \tag{22}$$

The third step entails evaluating the cost function. To accomplish this, we encode our problem into a cost function $C(\theta)$ as a linear combination of Pauli operators, executed on a quantum system.

In the fourth step, the evaluations are processed on a classical computer, where a classical optimizer reviews the results and determines the next set of values for the variational parameters. If an optimal solution already exists, it can be used as the starting point θ_0 for the optimization process. Using this initial state $|\psi(\theta_0)\rangle$ can help the optimizer achieve faster convergence.

The entire process is repeated until the convergence criteria of the classical optimizer are met, resulting in an optimal set of parameter values θ^* . The solution we obtain upon convergence of the optimizer will be $|\psi(\theta^*)\rangle = U_A(\theta^*)|0\rangle$.

Finally, its pseudocode can be found in Algorithm 5 (see Appendix C).

4.2.2 Quantum Support Vector Machines. QSVM builds upon the traditional **Support Vector Machine (SVM)** framework by incorporating quantum computing principles. Classically, SVMs operate by identifying a hyperplane in a high-dimensional feature space that effectively separates different data classes. The optimal hyperplane maximizes the distance, or margin, from the nearest points of each class, known as support vectors.

In contrast, the QSVM avoids the computationally intensive task of constructing a high-dimensional feature space by using a quantum circuit to embed data into a quantum feature space. This process, known as the quantum feature map, utilizes the parallel processing capabilities of quantum computers. The embedded states, denoted as $|\phi(\vec{x})\rangle$, reside in a Hilbert space of exponentially large dimension, enabling expressive representations that are often inaccessible classically.

Within this quantum feature space, the QSVM computes kernel functions, essential elements in SVMs that assess the similarity between pairs of data points. In quantum computing, these kernel functions are linked to the concept of fidelity, a measure that evaluates the similarity or “distance” between quantum states represented as state vectors or density matrices. Fidelity ranges from 0 (completely distinguishable states) to 1 (identical states). The quantum kernel is defined mathematically as

$$K_{ij} = |\langle \phi(\vec{x}_i) | \phi(\vec{x}_j) \rangle|^2, \tag{23}$$

where K_{ij} is the kernel matrix, $\vec{x}_i, \vec{x}_j$ are input vectors, and $\phi(\vec{x})$ is the quantum feature map. Fidelity values range from 0 (orthogonal states) to 1 (identical states), measuring the overlap between quantum states.

This is referred to as the quantum kernel trick, and allows QSVM to evaluate similarity in a quantum feature space without having to compute explicit feature vectors. The main computational advantage of this method over classical SVMs lies in the ability to efficiently estimate kernel values in exponentially large Hilbert spaces, which would be difficult to compute or even represent. Instead of explicitly mapping data to a high-dimensional space and calculating inner products, the quantum device does so implicitly through state superpositions, taking advantage of quantum parallelism and interference. This reduces the memory and computational overhead associated with traditional kernel methods and opens up the possibility of solving sorting problems that would otherwise be inaccessible to classical algorithms.

In practice, this overlap is estimated using circuits such as the SWAP test or the overlap test. In the SWAP test, a qubit ancilla mediates a controlled exchange between two quantum states, and the measurement of the ancilla provides information about the fidelity. The probability $P(0)$ of measuring the ancilla in the $|0\rangle$ state is related to the kernel value via Equation (24).

$$|\langle \phi(\vec{x}_i) | \phi(\vec{x}_j) \rangle|^2 = 2P(0) - 1. \quad (24)$$

This allows the construction of the full kernel matrix for SVM optimization. The resulting decision function takes the form of Equation (25).

$$f(\mathbf{x}) = \sum_i \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b, \quad (25)$$

where α_i are the learned coefficients from the SVM training process, $y_i \in \{-1, 1\}$ are the class labels of the training data, $K(\mathbf{x}_i, \mathbf{x})$ is the quantum kernel between a training point $\mathbf{x}_i$ and a test point $\mathbf{x}$, and b is the bias term.

Although QSVMs offer the potential for exponential gains, their performance is sensitive to noise and errors in today's quantum hardware. Kernel estimation is especially vulnerable: small gate or decoherence errors can distort fidelity values, producing unreliable or even non-positive semidefinite kernel matrices. Moreover, in high-dimensional quantum spaces, most pairs of states are nearly orthogonal, which can cause all kernel values to converge to zero and render classification inefficient.

To address this, several mitigation techniques are employed. One approach is to restrict the expressivity of the quantum feature map, thereby limiting the exploration of the full Hilbert space and preventing the generation of nearly orthogonal quantum states. Another strategy involves scaling the input data, which increases the average overlap between encoded states and improves the stability of kernel estimation. Additionally, projected or structured kernels can be used to simplify state comparisons by focusing on specific observables or subspaces, thus reducing sensitivity to noise and enhancing the discriminative power of the kernel matrix.

While the QSVM promises enhanced speed and efficiency for complex, high-dimensional classification tasks, its practical application remains challenging due to limitations in current quantum technology, such as qubit availability and quantum errors. Realizing the full potential of QSVM depends on advances in quantum computing, particularly in quantum error correction, an ongoing area of intense research.

Furthermore, there exist other versions of QSVM, like the PegasosQSVM. The PegasosQSVM algorithm is basically the same as QSVM, except that it is adapted for larger datasets and implements the Pegasus (Proximal Gradient Descent for Sparse Learning) algorithm to handle regularization terms that induce sparsity in solutions. This adaptation allows PegasosQSVM to be more efficient and scalable, especially in contexts where data is massive and sparsity is desired to improve interpretability and reduce model overfitting. There are also other types of kernels, which can be trained to achieve an optimised kernel, and therefore, better results can be achieved.

Finally, its pseudocode can be found in Algorithm 6 (see Appendix C).

4.2.3 Quantum Neural Networks. Classical neural networks are algorithmic models based on a series of interconnected nodes, or neurons, organized into multiple layers, with hyperparameters that can be adjusted using various techniques.

The concept behind QML is to merge the knowledge of quantum computing with classical machine learning to develop new and improved learning algorithms. QNNs implement this general principle by combining classical neural networks with parameterized quantum circuits. Because they lie at the intersection of these two fields, QNNs can be viewed from two different perspectives.

From a machine learning perspective, QNNs are algorithmic models that can be trained to find hidden patterns in data, much like their classical counterparts. These models can load classical

data (inputs) into a quantum state and then process it using quantum gates parameterized by trainable weights. From a quantum computing perspective, QNNs are quantum algorithms based on parameterized quantum circuits that can be trained variationally using classical optimizers.

In Figure 5 (see Appendix B), the general structure of the QNN can be observed. The first part of the network involves the data loading process. During this stage, classical data is transformed into its quantum version through a feature map. This transformation is crucial as it allows the data to be processed in a quantum state.

The second part of the network pertains to the data processing stage. Here, an ansatz circuit is used to determine the optimal weights for the network. As previously explained, this circuit acts as an optimizer that converges upon finding the appropriate weights for the network. This optimization process is fundamental for improving the accuracy and efficiency of the model. The ansatz is typically a parameterized quantum circuit that consists of a sequence of quantum gates dependent on trainable variables. Its architecture must be carefully designed to ensure a balance between expressivity and trainability, as poor choices can lead to issues such as the barren plateau problem, where the gradient of the cost function vanishes exponentially with the number of qubits. This phenomenon prevents the optimizer from making progress during training, as the parameter landscape becomes effectively flat. To mitigate this, several strategies have been proposed, such as using problem-inspired ansatz, initializing parameters near zero, layer-wise training, employing local cost functions instead of global ones, or leveraging adaptive circuit architectures. These approaches aim at maintaining meaningful gradients and improve the efficiency of the training process.

The overall parameterized quantum state produced by a QNN can be represented as Equation (26).

$$|\psi(\theta)\rangle = U_\theta U_\phi |0\rangle, \quad (26)$$

where U_ϕ is the encoding circuit (feature map), U_θ is the variational (ansatz) circuit, and $|0\rangle$ is the initial state of the quantum register.

Finally, in the last stage, the quantum state of the network is measured to analyze the results obtained after training. Measuring these quantum states allows us to evaluate the network's performance and determine if the learning objectives have been met. This final step is essential for validating the model's effectiveness and making necessary adjustments. The outcomes of the measurement correspond to expectation values of certain observables, which are used to compute a loss function. This loss is then minimized using a classical optimization routine, typically executed on a classical computer as part of a hybrid quantum-classical loop. The nature of the measurement also introduces statistical noise due to the probabilistic nature of quantum mechanics, so multiple repetitions (shots) are required to obtain reliable estimates. Managing this noise is a key practical challenge in current implementations.

The minimized loss function in a QNN is typically defined as the expected value of an observable O and is shown by Equation (27).

$$L(\theta) = \langle \psi(\theta) | O | \psi(\theta) \rangle, \quad (27)$$

where $|\psi(\theta)\rangle$ is the QNN state defined in Equation (26), and O is a Hermitian operator corresponding to the measurement observable related to the classification or regression task.

Compared to classical neural networks, QNNs offer several potential advantages. First, quantum systems operate in exponentially large Hilbert spaces, which may allow QNNs to represent and manipulate data in ways that are intractable for classical models. Second, they exploit quantum phenomena such as superposition, entanglement, and interference, which could provide more expressive hypothesis spaces for learning complex patterns. Third, in some specific contexts, such as learning quantum data distributions or modeling quantum systems, QNNs may achieve a quantum

advantage in efficiency or accuracy. In addition, they may require fewer parameters to approximate certain functions compared to deep classical architectures, which could reduce training complexity under the right conditions.

Finally, its pseudocode can be found in Algorithm 7 (see Appendix C).

4.3 Quantum Deep Learning Algorithms

As QML evolves, growing attention is being devoted to **quantum deep learning (QDL)** algorithms [121]. Unlike early QML models based on linear classifiers or shallow circuits, QDL approaches explore complex designs such as convolutional, recurrent, generative, and probabilistic networks. These are not mere quantum translations, but often require fundamentally new implementations adapted to quantum constraints. We review four representative QDL families: **Quantum Convolutional Neural Networks (QCNN)**, **Quantum Long Short-Term Memory Networks (QLSTM)**, **Quantum Generative Adversarial Networks (QGAN)**, and **Quantum Boltzmann Machines (QBM)**.

4.3.1 Quantum Convolutional Neural Networks. QCNNs [26] are quantum circuit architectures inspired by classical CNNs, designed to perform hierarchical feature extraction on quantum states. Introduced in the context of quantum phase recognition, QCNNs use alternating layers of convolution and pooling operations to compress and analyze input states. Each convolution layer applies unitary gates to local qubit groups, while pooling discards qubits or applies conditional operations, reducing dimensionality and introducing nonlinearity.

QCNNs are advantageous for near-term quantum processors due to their logarithmic circuit depth, which helps mitigate decoherence and hardware limitations in NISQ devices. Their translation invariance and robustness to noise further support applications in quantum error correction and image recognition. Though originally used for classifying quantum states, QCNNs can process classical inputs via appropriate encoding, extending their use to signal processing and medical imaging. Their efficiency and architectural simplicity offer advantages over unstructured variational circuits, especially in constrained hardware environments.

Finally, its pseudocode can be found in Algorithm 8 (see Appendix C).

4.3.2 Quantum Long Short-term Memory Networks. QLSTM networks generalize classical LSTMs by encoding input, hidden, and cell states in quantum registers processed through parameterized circuits acting as input, forget, and output gates. These gates evolve internal memory over time using VQC, with updates defined by quantum analogs of classical LSTM equations. The QLSTM operates in a hybrid loop, where quantum measurements guide classical optimizers to minimize prediction error. Encoding memory in quantum states allows superpositions and entanglement across time steps, enabling richer temporal modeling and potentially alleviating vanishing gradient issues in deep recurrent architectures.

A standard QLSTM involves a sequence of time steps in which input data is encoded into quantum states and interacts with quantum representations of prior hidden and cell states. The evolution is governed by quantum versions of the input, forget, and output gates, each implemented via a parameterized circuit. Specifically, the forget gate is $f_t = U_{\theta_f}(x_t, h_{t-1})$, the input gate is $i_t = U_{\theta_i}(x_t, h_{t-1})$, the candidate cell update is $\tilde{c}_t = U_{\theta_c}(x_t, h_{t-1})$, and the output gate is $o_t = U_{\theta_o}(x_t, h_{t-1})$. The internal memory is then updated as in classical LSTMs: $c_t = f_t \cdot c_{t-1} + i_t \cdot \tilde{c}_t$, and $h_t = o_t \cdot \tanh(c_t)$. These updates preserve the core structure of LSTM dynamics while operating on quantum-encoded representations.

QLSTMs are well-suited to sequential data tasks like time-series forecasting, natural language understanding, or quantum-enhanced signal processing. Despite their promise, QLSTMs face challenges such as coherence loss, circuit depth, and qubit resets. To mitigate these issues, unfolded designs using fixed-depth modules have been proposed [22].

Finally, its pseudocode can be found in Algorithm 9 (see Appendix C).

4.3.3 Quantum Generative Adversarial Networks. QGANs generalize the classical GAN framework to the quantum setting by pairing a parameterized quantum generator with a discriminator—either quantum or classical—in a two-player minimax game. The generator seeks to produce quantum states that approximate an unknown data distribution, while the discriminator learns to distinguish between real samples and those generated. Like classical GANs, QGANs rely on iterative adversarial training, but leverage quantum resources such as superposition and entanglement to encode and manipulate data distributions.

These models are typically implemented within a **variational quantum algorithm (VQA)** framework, where both the generator and discriminator are expressed as parameterized quantum circuits. Optimization proceeds via a hybrid loop, using quantum measurements to estimate loss functions and classical optimizers to update circuit parameters. This architecture enables QGANs to operate in NISQ environments and to tackle tasks involving either quantum or classical data.

Formally, the generator is implemented as a parameterized quantum circuit $U_G(\theta)$ applied to an initial product state, producing a generated quantum state that approximates the data distribution. The discriminator, either a quantum circuit $U_D(\phi)$ or a classical neural network acting on measured outputs, evaluates whether a given sample is real or fake. Training proceeds over T steps, where in each iteration a batch of real data samples $\{x_i\}$ is drawn from the dataset $\mathcal{D}$, and the generator produces a batch of fake states $\{z_i\}$. The discriminator loss is computed as

$$L_D = -\frac{1}{2} \sum_i [\log D(x_i) + \log(1 - D(z_i))], \quad (28)$$

where $D(\cdot)$ is the discriminator function that outputs the probability of an input being real. The discriminator parameters ϕ are updated to minimize L_D , improving its ability to distinguish real from fake data. Next, the generator produces a new batch $\{z'_i\}$, and its loss is given by

$$L_G = -\sum_i \log D(z'_i). \quad (29)$$

Minimizing L_G updates the generator parameters θ to increase the chance that fake data is classified as real. This adversarial process continues until the generator and discriminator reach a balance, known as Nash equilibrium. In QGANs, gradients are typically estimated using the parameter-shift rule or stochastic approximation, and optimization is performed via classical algorithms.

Finally, its pseudocode can be found in Algorithm 10 (see Appendix C).

4.3.4 Quantum Boltzmann Machines. QBMs are the quantum generalization of classical Boltzmann Machines, designed to model complex probability distributions through quantum principles such as superposition, entanglement, and quantum thermalization. While classical BMs represent probability distributions via stochastic neural networks trained through sampling, QBMs describe the target distribution as a quantum thermal state derived from a parameterized Hamiltonian. This formalism allows QBMs to capture correlations and distributions that are intractable for classical models.

Mathematically, a QBM is governed by a Hamiltonian $H(\lambda) = H_Z + H_X$, where H_Z encodes classical energy terms (e.g., pairwise interactions between visible and hidden nodes), and H_X introduces quantum fluctuations via off-diagonal elements, typically modeled as transverse fields. The model defines a quantum Gibbs state:

$$\rho_\lambda = \frac{e^{-\beta H(\lambda)}}{\text{Tr}(e^{-\beta H(\lambda)})}, \quad (30)$$

where β is the inverse temperature controlling the distribution's sharpness. Training proceeds over T iterations, where expectation values of relevant observables (e.g., $\langle Z_i \rangle$, $\langle Z_i Z_j \rangle$) are estimated both from the dataset $\mathcal{D}$ and the model distribution ρ_λ . The gradient for each parameter λ_k is calculated as

$$\partial_k = \langle O_k \rangle_{\text{data}} - \langle O_k \rangle_p, \quad (31)$$

and used to update the parameters as follows:

$$\lambda_k \leftarrow \lambda_k + \eta \cdot \partial_k. \quad (32)$$

This process aims at aligning the model's thermal state distribution with the statistics of the training data. Unlike classical Boltzmann Machines, which rely on sampling methods like Gibbs or contrastive divergence, QBMs require access to quantum thermal states, which can be prepared using quantum annealing [99] or variational thermalization techniques [35]. Their ability to capture quantum correlations makes them suitable for modeling physical systems and learning from quantum data [90].

Finally, its pseudocode can be found in Algorithm 11 (see Appendix C).

4.4 Quantum Reinforcement Learning Algorithms

Quantum reinforcement learning (QRL) explores how quantum resources, such as superposition, entanglement, and quantum parallelism, can enhance the ability of agents to learn from interaction with an environment. The field is still developing, but QRL shows promise in improving scalability, generalization, and sampling efficiency, especially in large or quantum-native environments. The following sections review two representative approaches: **Quantum Q-Learning (QQL)** and **Quantum Policy Gradient (QPG)**.

4.4.1 Quantum Q-learning. QQL is a quantum-enhanced variant of classical Q-learning, one of the foundational algorithms in reinforcement learning. Its goal is to learn an optimal action-value function $Q(s, a)$ that estimates the expected cumulative reward of taking action a in state s and following the learned policy thereafter. Unlike the classical approach, where $Q(s, a)$ is stored in a table or approximated by a neural network, QQL encodes this function within the parameters of a quantum circuit, enabling compact representation and potentially more efficient exploration of high-dimensional state-action spaces through quantum parallelism.

A typical QQL setup associates each state-action pair (s, a) with a parameterized quantum circuit $U_Q(\theta_{s,a})$. This circuit is initialized in a standard state (e.g., $|0\rangle^{\otimes n}$) and evolved through a variational ansatz. Upon measurement, the output is mapped to a Q-value estimate for that pair. During training, the agent interacts with the environment over T episodes. At each step, it follows an ϵ -greedy policy: with probability ϵ it selects a random action for exploration, and otherwise chooses $a = \arg \max_a Q_\theta(s, a)$. After executing the action, the agent observes a reward r and the next state s' .

The Q-value is then updated using the **temporal-difference (TD)** rule:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right], \quad (33)$$

where α is the learning rate and γ the discount factor. In the quantum setting, the target value $y = r + \gamma \cdot \max_{a'} Q_\theta(s', a')$ is computed classically, while the current Q-value $Q_\theta(s, a)$ is estimated from the quantum circuit. The loss function minimized at each iteration is

$$L = (Q_\theta(s, a) - y)^2. \quad (34)$$

This loss is used to update the circuit parameters θ via gradient descent, typically using the parameter-shift rule or other hybrid-compatible estimators. The variational formulation of QQL

enables it to be deployed on current NISQ hardware, provided the circuit depth is manageable and gradients can be reliably estimated.

Finally, its pseudocode can be found in Algorithm 12 (see Appendix C).

4.4.2 Quantum Policy Gradient. QPG extends the classical policy gradient method to quantum settings by encoding the policy in a **parameterized quantum circuit (PQC)**. This approach is well-suited for reinforcement learning tasks in continuous or high-dimensional action spaces, where classical value-based methods become inefficient. In QPG, the agent samples actions from a quantum distribution and adjusts the parameters of the PQC to maximize expected returns.

Let $\pi_\theta(a|s)$ denote the stochastic policy encoded by a quantum circuit $U(\theta)$. At each decision step, a quantum state $|\psi(\theta)\rangle$ is prepared and measured, and the resulting bitstring determines the selected action a . The parameters θ are optimized using gradient ascent over the expected cumulative reward $J(\theta)$, estimated through interaction with the environment. The policy gradient is given by

$$\nabla_\theta J(\theta) = \mathbb{E}_{s \sim \rho^\pi, a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a|s) \cdot Q^\pi(s, a)]. \quad (35)$$

In practice, the agent interacts with the environment over T episodes. For each episode, it collects a trajectory of state-action-reward tuples (s_t, a_t, r_t) . The return from each time step t is computed as

$$R_t = \sum_{k=t}^T \gamma^{k-t} r_k, \quad (36)$$

where γ is the discount factor. The gradient of the log-likelihood, $\nabla_\theta \log \pi_\theta(a_t|s_t)$, is estimated from the output of the quantum circuit, and the parameters are updated using:

$$\theta \leftarrow \theta + \eta \cdot R_t \cdot \nabla_\theta \log \pi_\theta(a_t|s_t), \quad (37)$$

where η is the learning rate. This update rule increases the probability of selecting actions that lead to higher long-term rewards. Gradient estimation can be performed via the parameter-shift rule, which is compatible with current quantum hardware and does not require full knowledge of the quantum state.

Finally, its pseudocode can be found in Algorithm 13 (see Appendix C).

4.5 Quantum Optimization Algorithms

Optimization is a key component in training quantum machine learning models, including QML, QDL, and QRL. These models require efficient strategies for tuning variational circuit parameters in high-dimensional and often nonconvex landscapes. Although there are some promising optimization algorithms, such as Variational Quantum Eigensolver [19] or Quantum Gradient Descent [102], the **Quantum Approximate Optimization Algorithm (QAOA)** stands out.

Thus, the QAOA is designed to find approximate solutions to combinatorial optimization problems, typically NP-hard. It was originally introduced to address problems that can be formulated as finding the bit string $z \in \{0, 1\}^n$ that maximizes a given cost function $C(z)$, usually encoded as a diagonal operator.

QAOA works by defining a parameterized quantum circuit composed of alternating layers of unitary operations derived from two Hamiltonians: a problem Hamiltonian H_C , which encodes the objective function to be optimized, and a mixer Hamiltonian H_M , which allows exploration of the solution space. The algorithm starts from an initial uniform superposition state and applies a sequence of these alternating unitaries, each parameterized by angles $\gamma = (\gamma_1, \dots, \gamma_p)$ and $\beta = (\beta_1, \dots, \beta_p)$ for a given depth p . The resulting quantum state is then measured to obtain candidate solutions, and the expected value of the cost function is used to guide the classical optimization of the (γ, β) parameters.

The QAOA state at depth p is given by Equation (38).

$$\left| \psi(\gamma, \beta) \right\rangle = \prod_{j=1}^p e^{-i\beta_j H_M} e^{-i\gamma_j H_C} \left| + \right\rangle^{\otimes n}, \quad (38)$$

where $|\psi(\gamma, \beta)\rangle$ denotes the QAOA state after p layers, with each layer composed of alternating unitaries generated by the problem Hamiltonian H_C and the mixer Hamiltonian H_M , parameterized respectively by angles γ_j and β_j . The initial state $|+\rangle^{\otimes n}$ is the uniform superposition over all computational basis states of n qubits.

This state encodes a distribution over bit strings that depends on the angles γ and β . The goal of the classical optimizer is to find the angles that minimize (or maximize) the expectation value of the cost Hamiltonian, as seen in Equation (39):

$$\langle H_C \rangle = \langle \psi(\gamma, \beta) | H_C | \psi(\gamma, \beta) \rangle. \quad (39)$$

The efficiency of QAOA depends on the number of layers p . As p increases, the algorithm can converge to the optimal solution, although this also increases the circuit depth and susceptibility to noise. One of the main strengths of QAOA lies in its flexibility, as it can be adapted to different types of constraint satisfaction problems and extended with problem-specific custom mixers or ansatz.

Although a general quantum speedup remains unproven, QAOA is widely used for quantum optimization. Furthermore, it is widely supported by software frameworks such as Qiskit, PennyLane, and Cirq, which provide tools for building circuits, simulating noise models, and running on quantum hardware.

Finally, its pseudocode can be found in Algorithm 14 (see Appendix C).

5 Hybrid Architectures

Hybrid quantum-classical architectures have emerged as a central design strategy in QML, particularly in the NISQ era. These models combine expressive quantum circuits with classical components, making the most of both to overcome current hardware limitations. Hybrids allow shallow quantum models to remain trainable and scalable, enabling practical implementations of variational learning. De Luca et al. [27] provide a broad overview of these approaches, highlighting their accessibility through cloud platforms and their application in machine learning. The work also introduces pedagogical tools for quantum gates and reviews a growing number of hybrid models deployed on real quantum hardware, emphasizing their importance in bridging near-term limitations and long-term quantum advantage.

Hybrid models can be broadly categorized based on the direction and depth of information flow between quantum and classical components. Two main types are commonly identified in the literature: (1) classical-to-quantum (CQ), where classical preprocessing feeds into quantum models, and (2) quantum-to-classical (QC), where quantum circuits perform feature extraction and classical components handle the final predictions. Fully hybrid models (FQC), with alternating quantum and classical layers in a joint optimization loop, are typically considered an advanced subtype of QC architectures.

CQ models apply classical preprocessing, such as convolutional filters or dimensionality reduction, before encoding data into quantum states. This approach is particularly effective when classical stages extract structured features from large-scale inputs, allowing quantum circuits to focus on expressive transformations within hardware constraints. Kim et al. [57] explored this idea through a transfer learning scheme, where a classical CNN trained on Fashion-MNIST provided weights to a QCNN trained on MNIST, achieving strong results with shallow quantum circuits. Wang et al. [122] proposed SHQCNN, a hybrid model addressing gradient vanishing in deep QNNs by combining

classical preprocessing with a variational quantum layer and mini-batch training. Their model achieved 99.58% accuracy, outperforming its classical counterpart and demonstrating enhanced robustness on noisy data.

QC models use quantum circuits as feature extractors, followed by classical layers that perform prediction. In this setup, quantum expectation values serve as intermediate representations, which are passed to classical neural networks or classifiers. This architecture balances quantum expressiveness with classical stability and has proven effective in various domains. For example, Dou et al. [31] proposed a hybrid convolutional structure where parameterized quantum circuits act as sliding kernels, achieving strong performance on subsets of MNIST. Xiang et al. [125] integrated quantum layers into CNNs for breast cancer diagnosis, showing improved generalization across large-scale medical datasets. Similarly, Hafeez et al. [6] designed a compact H-QNN that combines a two-qubit circuit with a classical backbone, reporting 90.1% accuracy and robust behavior in low-data settings.

FQC architectures alternate quantum and classical components within a single learning loop, optimizing both jointly. This deep integration is particularly suited to complex tasks like generative modeling and reinforcement learning, where both adaptivity and expressive power are crucial. Chen et al. [21] embedded a VQC into a deep Q-learning agent for maze navigation, showing competitive performance and efficient training with fewer parameters than classical counterparts. In a generative context, Liu et al. [70] introduced ILL-QGAN, a hybrid adversarial model that progressively grows the circuit using layer-wise legacy learning. Their method improved convergence and accuracy over standard QGANs, while maintaining compatibility with NISQ-era hardware.

It is worth mentioning that hybrid models face major challenges. Vanishing gradients (known as barren plateaus) limit trainability in deep or highly entangled circuits [77]. Simulation costs grow with batch size and circuit complexity [5], and scalability is constrained by limited parallelism and incompatibilities among simulators, backends, or autodiff tools. Nonetheless, frameworks like PennyLane [8] and Qiskit [48] are improving integration across classical and quantum layers.

6 Quantum Frameworks, Cloud Integration, and Datasets

This section reviews the existing frameworks for QML (Section 6.1), discusses the integration of QML with cloud-based quantum platforms (Section 6.2) and introduces existing datasets (Section 6.3).

6.1 Existing Frameworks

Various frameworks and libraries enrich the landscape of quantum computing, each offering distinctive features and capabilities for developing and executing quantum algorithms. Given the variety of quantum computing frameworks available, it is natural to consider which one is the most suitable for beginners. Each framework has its own strengths and weaknesses, which can significantly impact the learning experience and the types of projects you may want to pursue.

In this section, we provide an overview of several prominent quantum computing frameworks and libraries, including Qiskit, PennyLane, Q#, Cirq, Wolfram Quantum, Amazon Braket, Strawberry Fields, TensorFlow Quantum, and OpenFermion. We discuss each framework's architecture, features, and suitability for various quantum computing tasks. Additionally, we highlight the advantages and limitations of each framework to assist researchers and practitioners in selecting the most appropriate tool for their specific needs.

Qiskit [48], developed by IBM, is an open-source quantum computing framework that provides tools for quantum circuit construction, simulation, and execution, with a user-friendly programming interface and extensive documentation. Qiskit supports hybrid quantum-classical computing and integrates with classical machine learning libraries, though it may face scalability and resource utilization limitations.

PennyLane [8], developed by Xanadu, focuses on differentiable programming and hybrid quantum-classical computing, offering seamless integration with TensorFlow and PyTorch. It excels in optimization techniques and supports various quantum hardware platforms, but it has a steeper learning curve and a smaller ecosystem.

Q# [78], developed by Microsoft, provides a high-level programming model for quantum algorithms, integrating with C# and Python, and offers strong support for quantum simulation and debugging tools, though its ecosystem is smaller.

Cirq [38], an open-source framework from Google, allows for the design, simulation, and execution of quantum circuits, offering flexibility and a modular architecture despite a steeper learning curve and potential lack of tooling and visualization capabilities.

Wolfram Quantum [123], part of the Wolfram Language ecosystem, provides tools for symbolic quantum computation, circuit simulation, and algorithm development, with high-level symbolic manipulation capabilities, though it may have limited support for quantum hardware interfaces.

Amazon Braket [3], an AWS service, offers access to quantum hardware from multiple vendors and provides a development environment for quantum algorithms, integrating seamlessly with AWS infrastructure, but may be perceived as a new platform with evolving features and potentially prohibitive pricing.

Strawberry Fields [56], from Xanadu, focuses on continuous-variable quantum computing and integrates with PennyLane, making it suitable for quantum photonic systems, though it may be limited for discrete-variable tasks. TensorFlow Quantum [13], developed by Google and others, enables the construction of hybrid quantum-classical models using TensorFlow, offering scalable and efficient machine learning workflows but requiring familiarity with both TensorFlow and quantum computing concepts.

TorchQuantum [126] is a recent library developed by Tsinghua University, specifically designed for QDL tasks. It integrates PyTorch with parameterized quantum circuits, allowing the training of quantum models using classical techniques such as self-difference. It is especially useful for experienced PyTorch researchers interested in QML.

IBM Quantum Lab [47] is an interactive cloud platform that provides direct access to IBM quantum hardware, integration with Qiskit, and visual tools for developing and running quantum algorithms. It's an excellent practical gateway for both beginners and advanced users who want to experiment with real-world quantum computing. Table 3 (see Appendix A) summarizes the quantum computing frameworks discussed before, highlighting key attributes such as the company behind the development, licensing structure, programming languages supported, and the year of their introduction.

Choosing the right framework can feel challenging, especially for those who are new to quantum computing, as each option offers distinct features that cater to different interests and skill levels. Table 4 (see Appendix A) presents a concise overview of the various frameworks, highlighting their primary benefits and key considerations in an attempt to help researchers to make the right choice. This table might be a reference point for selecting a framework that aligns with the goals and background. Comparing aspects such as ease of use, community support, and integration capabilities may facilitate the initial foray into QML.

Consequently, if someone is starting out and looking for a general framework to learn the fundamentals of quantum computing, Qiskit is often recommended due to its documentation and supportive community. It allows you to explore various quantum computing concepts and applications effectively. However, if someone is interested in integrating quantum computing with machine learning immediately, PennyLane or TensorFlow Quantum could also be excellent choices. Finally, Table 5 (see Appendix A) summarizes the support each framework offers for the main QML algorithms analyzed. The fact that an algorithm is not marked does not imply that it cannot be

implemented with a given framework, but rather that the framework does not provide built-in classes or high-level methods that facilitate its direct implementation.

6.2 Integration of QML with Cloud-based Quantum Computing Platforms

In the last years, cloud-accessible quantum computers have grown from experimental prototypes to fully managed services capable of running millions of circuit evaluations per day. Public platforms such as IBM Quantum, Amazon Braket, Google Quantum AI and Microsoft Azure Quantum now offer researchers immediate and uniform access to superconducting, trapped ion and neutral atom processors, abstracting from the underlying cryogenic hardware and vendor-specific toolchains. Allowing the same QML experiment to run on heterogeneous back-ends under identical calibration snapshots has made these services indispensable for reproducible QML research. They facilitate benchmarking between technologies, provide calibrated noise data for error mitigation studies, and extend parameter sweeps far beyond the capacity of any local test. The main cloud platforms and their specific QML integration points are listed in Table 6 (see Appendix A).

6.3 Quantum Datasets

Quantum datasets provide a consistent basis for evaluating and comparing quantum algorithms and hybrid models. However, preparing quantum-compatible datasets is often complex, as it requires either preprocessing classical data or generating data through simulations or quantum hardware.

A notable example is *QDataSet* [88], which offers a scalable framework for converting classical datasets into quantum circuit representations. Each dataset includes Qiskit-compatible circuits that encode data using methods such as angle or amplitude encoding, along with standardized splits, labels, and metadata describing the encoding process. This structure supports reproducibility and simplifies benchmarking by allowing researchers to focus on model design rather than data preparation. *QDataSet* covers a range of machine learning tasks, including classification, regression, and generative modeling, using both classical benchmarks (e.g., Iris, MNIST) and synthetic datasets tailored to quantum applications.

The *PennyLane Quantum Data* module [87] offers a collection of datasets designed for use in QML workflows. Examples include the moon dataset for binary classification and several molecular datasets intended for quantum chemistry simulations using variational algorithms. Although relatively small, these datasets are practical for experimentation, benchmarking, and instructional use within hybrid quantum-classical settings.

The *Open Quantum Data Commons (OpenQDC)* [79] is a large-scale repository of quantum mechanics datasets. It includes nearly 40 datasets covering over 1.5 billion molecular geometries across 70 atomic species and 250 quantum methods. Each dataset is standardized and enriched with metadata and is accessible through a Python API that supports efficient formats.

Table 7 (see Appendix A) summarizes the main features of the datasets introduced in this section.

7 Quantum Machine Learning Current Applications

Current research in QML can be broadly divided into two primary approaches: theoretical and practical. Theoretical studies focus on the conceptual development of methodologies, including discussions, frameworks, and guidelines for advancing the field. In contrast, practical studies emphasize real-world applications through performance evaluations, comparative analyses, and implementations. Accordingly, Section 7.1 provides a detailed review of theoretical advancements, while Section 7.2 examines practical contributions to QML.

7.1 QML Theoretical Proposals

The theoretical proposals address QML models or methodologies from their conceptual basis, offering mathematical definitions and prototypes but without practical implementations or tools.

These proposals serve as a starting point for long-term research, where subsequent efforts focus on prototyping, testing, and demonstrating the model's performance. Key objectives include proving efficiency gains, such as reduced training time, when using quantum machines compared to classical ones. However, these research endeavors face significant challenges, including the lack of quantum operating systems, memory, and data storage, all of which complicate the development of viable QML solutions.

On the other hand, implementing advanced quantum software presents its own difficulties. Quantum programming must adhere to paradigms such as quantum circuit programming or quantum adiabatic programming, both of which impose strict constraints that hinder the development of sophisticated software. In many ways, quantum programming today resembles the early days of classical computing, where systems were programmed with wires and plugs. Moreover, training and deploying QML models on actual quantum hardware remains a complex task, requiring novel and innovative solutions to overcome these limitations.

The following articles present quantum models from their mathematical foundation to explore theoretical proposals, serving as a starting point for future developments. Several studies establish the theoretical foundations of QML and analyze fundamental quantum algorithms. Jadhav et al. [50] outline the foundations of quantum computing, highlighting pioneering contributions such as Shor's and Grover's algorithms, which demonstrate the potential of quantum computing for specific tasks such as integer factorization and unstructured search. [76] and [44] explored how the integration of quantum computing with machine learning can overcome the limitations of classical methods, highlighting algorithms such as QSVMs and quantum k-means clustering, which show significant improvements in classification and clustering tasks.

Several studies focus on the development and optimization of specific QML algorithms. In [28], the authors introduced innovative quantum information processing methods that improve the efficiency of unsupervised tasks. Yang et al. [130] addressed noise in intermediate-scale quantum computers and proposed adaptive mechanisms to reduce noise requirements while maintaining privacy. [109] presented the **parametrized Hamiltonian learning (PHL)** model, optimizing the evolution of quantum systems for applications such as image segmentation. In addition, [52] and [16] explored parametrized quantum circuit models and quantum algorithms to improve the interpretability and efficiency of machine learning models.

QML is applied in multiple sectors, demonstrating its versatility and disruptive potential. Kumar et al. [51] investigated healthcare, finance, and automotive applications, highlighting how QML can outperform classical systems in solving complex problems. Rana et al. [101] and Li et al. [66] applied QSVM and QNN to handwritten character recognition and supervised classification tasks, respectively, showing improvements in accuracy and efficiency. [103] integrated ML with quantum key distribution protocols to optimize communication security, while the authors of [136], developed frameworks for blind QML, guaranteeing privacy in ML task delegation.

The practical implementation of QML on quantum hardware is an area of intense development. [96] and [65] reviewed quantum platforms such as IBM Q, IonQ and D-Wave, discussing architectures and tools such as Qiskit and TensorFlow Quantum for developing hybrid quantum-classical models. Koshka [60] and Alchieri et al. [2] discussed the differences between adiabatic and gate-based quantum computing, emphasizing the role of NISQ devices in the execution of QML algorithms and the need for robust techniques to handle noise and decoherence.

Despite advances, QML faces several challenges that must be addressed for widespread adoption. [74] and [112] highlighted the current limitations of quantum hardware, such as qubit stability and high data input costs, as well as the need for more efficient and robust algorithms in the face of noise. In addition, Abbas et al. [1] pointed out that, although quantum ensembles of classifiers have potential, there are difficulties in high-dimensional data sets. Future research should focus on

improving scalability, reducing susceptibility to noise, and developing theoretical frameworks that allow quantum advantages to be fully exploited in practical scenarios.

Integrating quantum methods with classical techniques and improving the explainability of models are crucial to developing QML. O’Quinn et al. [85] and Burge et al. [16] discussed how hybrid algorithms can benefit from the strengths of both paradigms, while [52] addressed the need for explainable models in the quantum context. These approaches seek to improve performance and ensure that quantum models are interpretable and transparent, which is fundamental for their adoption in critical applications.

7.2 QML Practical Proposals

Once we have finished reviewing the theoretical articles, we move on to the practical articles. Numerous studies explore the implementation of quantum algorithms for practical classification and regression tasks. Bhavsar et al. [9] proposed algorithms such as VQCA and PegasosQSVM for classifying potentially hazardous asteroids, achieving 98.11% accuracy. The authors of [118] developed an FCQ-CNN for the diagnosis of ischemic heart disease, improving efficiency and precision compared to classical methods.

Amine et al. [134] highlighted QNNs and VQCA, which outperform CNNs and classical classifiers in tasks such as MNIST and Iris. Saloni et al. [29] investigated QSVMs for earthquake prediction, demonstrating quantum advantages in handling non-linear patterns. Michele et al. [40] explored QSVMs for fraud detection, integrating hybrid quantum-classical methods to enhance feature selection.

The work [133] introduced QRAC to reduce the number of qubits needed in VQCs, while the study [46] proposed Meta-VQA, integrating RNNs and variational algorithms to optimize complex tasks. Blancea et al. [11] presented a hybrid VQCA for particle physics classification that exceeds classical models in accuracy.

Tailong et al. [127] applied QML for ghost imaging, achieving 10% higher accuracy in object identification than classical methods. Similarly, Pandey et al. [86] explored QML in NLP tasks, demonstrating improvements in QLSTM models for part-of-speech tagging across multilingual datasets.

QML has demonstrated a significant impact in critical fields. Raja et al. [100] proposed a system to diagnose retinopathy of prematurity, with QSVM achieving 95.5% accuracy. Sharanya et al. [94] employed QSVC and QNN to classify cardiovascular diseases, reaching 97.31% accuracy. In industrial prognostics, Caio et al. [113] used hybrid quantum-classical models for the analysis of vibration signals in rotating machinery, highlighting configurations with CZ entanglement gates.

Sebastianelli et al. [108] applied QCNN to satellite images, achieving a 10% improvement in land-use classification. Satpathy et al. [64] utilized QML to enhance energy forecasting in extreme IoT environments, achieving 98.10% accuracy.

In chemistry, Teppei et al. [115] demonstrated the ability of QML to outperform classical regression models in the prediction of toxicity. Hin Li et al. [67] explored Gaussian quantum kernels to enhance data representation in classification tasks.

In the NISQ era, optimizing resources is critical. Tobias et al. [43] proposed methods to measure quantum kernels, reducing computational costs and execution time. Nam et al. [83] introduced QES to optimize parameterized circuit architectures, while Nguyen et al. [24] presented Qsun, a quantum virtual machine that accelerates variational algorithms.

Phalak and Ghosh [91] developed adaptive shot optimization strategies to reduce training costs. Masahiro et al. [59] implemented quantum dropout, improving the generalization of the model. Additionally, Rohit et al. [30] proposed compression schemes to adapt classical data for NISQ devices.

Marco et al. [111] highlighted hybrid quantum-photonic networks for classification, achieving high accuracy with minimal parameters. Gong et al. [37] demonstrated that QCNN outperforms CNN in computationally intensive tasks such as MNIST, with advantages in convergence and accuracy.

Several studies evaluate the performance of hybrid models. Ricardo et al. [80] compared QSVM and QNN, showing 3-7% improvements in accuracy over classical methods. Sierra-Sosa et al. [110] investigated the impact of quantum state preparation on hybrid models, highlighting the advantages of amplitude encoding over angle encoding.

Chen et al. [23] explored variational circuits in reinforcement learning, achieving results comparable to classical methods. Andrés et al. [4] evaluated dimensionality reduction strategies in reinforcement learning, achieving greater cumulative rewards with compact quantum models.

Xiong et al. [128] proposed quantum inner product circuits with an efficient simulation scheme, enabling their integration into hybrid learning processes. Applied to tasks such as classification and K-Means clustering, the circuits maintained accuracy while reducing simulation time by over 68000 times.

Qu et al. [98] developed QMFND, a fake news detection model that uses amplitude encoding and QCNN to fuse text and image features. Tested on Gossip and Politifact, it outperformed classical baselines with great robustness to noise.

Li et al. [69] introduced EQNAS, a quantum neural architecture search method based on evolutionary strategies. This method improved the accuracy of QNN on MNIST and warship datasets while reducing the parameters by more than 20%. In [132], the authors present BQNAS, a method that combines supernet training with quantum evolution to improve search efficiency. The model achieved 97.27% accuracy on CIFAR-10 and 81.36% on CIFAR-100, outperforming classical NAS approaches.

The work [68] presents QEA-QCNN, which optimizes QCNN architectures through quantum evolution. The method reduces circuit complexity while enhancing expressiveness and facilitating deployment on NISQ devices.

Bikram et al. [54] reviewed Grover's Algorithm and quantum methods in tasks such as pattern recognition, while Shivani et al. [106] emphasized speed advantages of QSVM over classical SVM. Finally, Rodriguez et al. [105] compared Qiskit and PennyLane over hybrid architectures, providing insights into their advantages and limitations for quantum-enhanced machine learning.

Recent practical contributions also include interdisciplinary approaches combining quantum-inspired methods and classical AI techniques in financial applications. The study by [63] conducted a bibliometric analysis of 1890 journal articles related to artificial intelligence in business and corporate finance, highlighting the prevalence of artificial neural networks, deep learning and support vector machines in most research niches, while noting the limited use of techniques such as topic modelling and fuzzy neural networks. The authors also proposed a foundational framework for integrating AI into economic science, supported by a customised randomized Monte Carlo algorithm. In [7], the authors presented a **high-frequency rebalancing algorithm (HFRA)**, which combines pair trading with threshold-based rebalancing strategies and was tested on real cryptocurrency market data under different trends and volatility regimes. The study found that increasing portfolio size improves HFRA returns in high upside volatility environments and mitigates losses in downtrends. Compared to the periodic rebalancing and threshold strategies, the HFRA outperformed in the long run, statistically validated by the Fisher-Pitman permutation test.

Despite its promise, QML faces several challenges. Michele et al. [40] and Hiroshi et al. [133] underscored limitations in current hardware and encoding techniques. Mukta et al. [84] emphasized the need for robust quantum systems for complex tasks, while Rishabh et al. [41] highlighted scalability challenges in QNN for secure communications.

Payden et al. [14] explored quantum encodings for advanced materials, while Clouâtre et al. [25] proposed learning-enabled control methods for closed quantum systems. QML can outperform classical methods across various applications and its continued evolution on NISQ devices signals a promising future. However, realizing its full potential will require addressing existing challenges while continuing to refine its theoretical and practical underpinnings. Future research should prioritize scalable, noise-resilient models and the development of quantum methods that are both efficient and interpretable, helping QML's integration into real-world systems. Finally, Tables 8, 9, 10, and 11 (see Appendix A) provide an overview of all the reviewed articles. Tables 8 and 9 include columns for the reference of each article, the main topic addressed, and the type of machine learning task carried out. Similarly, Tables 10 and 11 contain these same three columns, along with an additional one specifying the programming framework used to implement the solution. The last three columns, common to all four tables, summarize key aspects of the experimental setup. Specifically, they also report the number of qubits employed during experimentation and the number of data samples used (*#Dataset*) or, in cases where a range is shown, it reflects the use of multiple datasets. The rightmost column identifies the type of noise present in the quantum execution environment, with three possible values: NFS for a noise-free simulator, NS for a noisy simulator, and ND for execution on a real quantum device.

8 Challenges and Future Work

QML presents substantial opportunities but also significant hurdles. This section identifies and critically examines key challenges, highlighting recent specific research cases and experimental results to provide detailed guidance for advancing future research.

One of the foremost challenges in QML is hardware constraints, primarily driven by qubit coherence times, limited qubit connectivity, and significant noise levels in current quantum devices. These constraints severely restrict the practical implementation of algorithms, particularly those requiring complex quantum circuits or deep quantum neural networks. Recent experimental results, such as Google's demonstration of quantum supremacy using their Sycamore processor [5], emphasize the need to optimize algorithms for near-term quantum devices. Similarly, IBM's experiments with error mitigation techniques, including zero-noise extrapolation [58], provide practical guidance on implementing error-resilient models on noisy quantum hardware. Future research should focus on developing quantum algorithms that explicitly consider current hardware constraints, employing hardware-adaptive ansatz and quantum error correction or mitigation techniques.

Another critical challenge is scalability. Quantum algorithms such as QAOA or QNN often experience barren plateaus, where the gradient vanishes exponentially with increasing circuit depth. Recent research by Grant et al. [39] experimentally verified this barren plateau phenomenon, highlighting the importance of efficient parameter initialization and layer-wise training approaches, which showed significant improvement in convergence and scalability. Future work should further explore structured ansatz designs, as demonstrated in experimental cases in [89], which achieved more scalable and trainable quantum neural networks by leveraging problem-specific ansatz structures.

The process of efficiently encoding classical data into quantum states remains a significant obstacle. Traditional embedding methods, such as amplitude encoding, require a prohibitively large number of qubits, whereas simpler encoding strategies like angle encoding may lead to insufficient data representational capacity. Recent advances, such as the data re-uploading strategy experimentally demonstrated by Pérez-Salinas et al. [97], show significant improvements in data representation within quantum circuits. Future research should investigate hybrid encoding techniques and their practical performance in tasks like quantum classification and quantum-enhanced image processing, using specific benchmarks.

Currently, many QML approaches lack interpretability and transparency, complicating their practical utility. Research in [45] experimentally demonstrated quantum advantage in certain supervised learning tasks, emphasizing the importance of identifying specific domains and conditions under which quantum models outperform classical methods. Future directions should systematically identify and experimentally validate these niche applications, providing clear interpretability and measurable quantum advantage criteria.

The ethical dimensions of quantum technology, including fairness, accountability, and security implications, remain underexplored in QML research. A promising direction highlighted by recent experiments in quantum-safe cryptography, such as the post-quantum cryptographic protocols experimentally validated by NIST competitions [82], illustrates the necessity of incorporating ethical considerations and security frameworks in QML development.

Hybrid quantum-classical methods represent one of the most promising research directions, as evidenced by recent experimental implementations such as the hybrid QGANs developed in [137]. Such methods take advantage of the best of quantum and classical worlds, compensating for quantum limitations and exploiting classical computational strengths. Future research should explore experimentally validated hybrid methods in more realistic and complex scenarios, including quantum-enhanced optimization, drug discovery, financial modeling, and natural language processing, systematically benchmarking against classical methods to identify practical quantum advantages. Finally, Table 12 (see Appendix A) includes a summary of the challenges found in the literature, highlighting the multifaceted nature of the obstacles that QML must overcome to reach its full potential.

9 Conclusions

QML is a rapidly advancing interdisciplinary field, revolutionizing computational paradigms by combining quantum mechanics with machine learning. This survey has presented a structured and thorough overview of QML, covering its foundational principles, algorithms, frameworks, datasets and diverse applications. By synthesizing more than 135 works, this work has highlighted the field's trajectory, achievements, and the pressing challenges it must address.

The foundational concepts of QML highlight its potential to surpass classical methods in tasks requiring high-dimensional data processing, optimization, and computational efficiency. Algorithms such as QSVM, VQC, and QNN demonstrate the applicability of QML across various domains, including healthcare, finance, and communication networks. Furthermore, advancements in frameworks and platforms, such as Qiskit and PennyLane, have enabled more accessible development and deployment of QML solutions.

Notable achievements in QML research, including improved classification accuracy and enhanced data modeling, illustrate its disruptive potential. Applications in critical fields such as cryptography, personalized medicine, and resource optimization further highlight the practical benefits of QML. However, challenges persist, particularly in hardware scalability, algorithmic design, and noise mitigation. Addressing these issues will require collaborative efforts across quantum computing, machine learning, and engineering disciplines.

Integrating hybrid quantum-classical algorithms and developing error-resilient models represent promising directions for QML. Continued investment in hardware innovation and the refinement of quantum algorithms is essential to unlocking the full potential of QML in solving real-world problems. As the field matures, its contributions are expected to transcend theoretical advances, driving meaningful impacts across industries and reshaping the computational landscape.

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